Business Analytics - Application Workflow & Architecture

1. System Architecture Overview

The application follows a clean, decoupled 3-tier architecture ensuring maintainability and scalability:

- **Frontend (Client-Side):** React 19, Vite, React Router, Axios
- **Backend (Server-Side):** Django 5.2, Django REST Framework, pandas, scikit-learn
- **Database:** SQLite (stores models, datasets metadata, prediction histories)

2. Complete Data Flow (Request Lifecycle)

When a user performs an action (e.g., uploading a dataset or running a prediction), the system follows this workflow:

Step 1: User Interaction (Frontend)

The user interacts with a React component (e.g., clicks "Upload" or "Predict"). The component collects the required data.

Step 2: API Call (Axios)

The frontend triggers a function in the `src/api/` directory (like `datasetApi.js`), which sends an HTTP POST/GET request to the Django backend.

Step 3: URL Routing & View Validation (Backend)

Django's `urls.py` routes the request to a specific function in `apps/analytics_api/views.py`. The view handles authentication and validates the incoming data parameters.

Step 4: Business Logic Orchestration (Services)

The view delegates the complex operations to the `services/` layer (e.g., `eda_service.py`, `prediction_service.py`). This keeps the views lightweight and focused solely on HTTP handling.

Step 5: Machine Learning & Core Processing (ML Layer)

If the task involves predictions or data cleaning, the service calls specific modules from the `ml/` directory:

- `preprocessing.py` (handling missing values, encoding)
- `regression.py / classification.py` (model building)
- `metrics.py` (evaluating accuracy/RÂ²)

Step 6: Database Persistence (Models)

The service queries or saves the results to the SQLite Database using Django Models (`Dataset`, `PredictionRun`, `GeneratedReport`).

Step 7: HTTP Response

The service returns the structured data back to the view, which packages it into a JSON Response. The frontend receives this response and dynamically updates the UI.

3. Backend Module Responsibilities

Module Directory	Primary Responsibility
apps/analytics_api/	Database Models, Serializers, URL routing, and lightweight Views (API endpoints).
services/	Core business logic separating concerns (Dataset, EDA, Visualization, Prediction, Reports).
ml/	Machine learning algorithms, data cleaning pipelines, and evaluation metrics.

core/	Reusable utilities and authentication mechanisms used across the project.
media/	Storage for uploaded CSVs, generated charts, and PDF reports.